2020 25th International Conference on Pattern Recognition (ICPR)
Milan, Italy, Jan 10-15, 2021

# Documents Counterfeit Detection Through a Deep Learning Approach

Darwin Saire
University of Campinas – UNICAMP
Institute of Computing
Campinas, SP – Brazil
Email: darwin.pilco@ic.unicamp.br

Salvatore Tabbone
Université de Lorraine, CNRS, LORIA
LORIA, UMR 7503
Campus Scientifique, 615 Rue du Jardin-Botanique,
F-54506 Vandœuvre-lès-Nancy, France
Email: salvatore.tabbone@univ-lorraine.fr

***Abstract*—The main topic of this work is on the detection of counterfeit documents and especially banknotes. We propose an end-to-end learning model using a deep learning approach based on Adapnet++ which manages feature extraction at multiple scale levels using several residual units. Unlike previous models based on regions of interest (ROI) and high-resolution documents, our network is feed with simple input images (i.e., a single patch) and we do not need high resolution images. Besides, discriminative regions can be visualized at different scales. Our network learns by itself which regions of interest predict the better results. Experimental results show that we are competitive compared with the state-of-the-art and our deep neural network has good ability to generalize and can be applied to other kind of documents like identity or administrative one.**

## I. Introduction

Forensic science is the application of scientific knowledge and methodology to legal problems and criminal investigations[1]. Forensic research in document examination is a hot topic still open. Due to the rapid development of new information technology and the wide use of the Internet, digital information is obtained easily. Digital content can be illegally manipulated for some malicious issues such as forged documents or banknotes counterfeit. Nowadays, with the digital technology available to each one it becomes available even to untrained users to make a fake document. Therefore, counterfeiting is an increasing problem for both government agencies and private companies. From an economical point of view it can cause substantial havoc.

In the field of image forensics, major research efforts have been made for security applications focused on image forgery detection and authentication [1]. Some works have inspired researches in documents forensic where efficient applications have been proposed for documents and banknotes counterfeiting detection. Particularly, most of the approaches are based on visual features and texture characteristics. Most of the cases, documents are forged applying cut and paste techniques to manipulate figures, letters, or words. In [2], authors use uniform Local Binary Patterns (LBP) to capture copy-paste or deletion operations through discontinuities in the texture. However, the approach needs some preprocessing to define patch (ROI) around each textual connected component and to remove the document. Other features are based on color and geometric features to identify false documents [3], [4].

Identifying the source printer of a document is another important purpose in forgery detection. The ability to discriminate between original documents and their photocopies is an important task since original digital content can be illegally manipulated and freely printed into documents. In this perspective, printer feature characteristics have been proposed. For instance, DCT coefficients [5] have been used as a fingerprint for the printing technology used to create the document. In [6], 15 statistical features are proposed to describe pattern noise produced by 20 flatbed scanners. The approach is subject to a fine segmentation between the foreground and the background (text segmentation and noise filtering). A deep learning approach based on Convolutional Neural Networks (CNNs) is proposed in [7] to learn automatically the best features which characterize the printer source. Reported results between a feature-based SVM and the proposed CNN classifiers show comparable classification performances for scanned documents but on microscopic input images, SVM is clearly better. Furthermore, an interesting comparative study on texture descriptors [8] to differentiate between genuine and counterfeit documents on several ROI textures show that performances of several texture hand-crafted descriptors are comparable to CNN descriptors.

Other works come back to the physical phenomena and focused on spectroscopic techniques for banknotes counterfeiting. Non-destructive analysis techniques are deployed to acquire information about the chemical composition of genuine and forged materials. Spectral analysis (for instance infrared or X-ray spectroscopy) allows to discriminate between genuine and counterfeit banknotes [9], [10]. Baek et al. [11] focused on a special sensor for ATM applications where RGB and infrared images are produced. A classifier is feeded with some local features defined on particular circular and rectangular regions.

Sensors that are capable to identify forged banknotes are rather expensive. Some methods are defined for non-controlled environment where a mobile device is used for documents acquisition [12]–[15]. Dimitti et al. [14] proposed the HOG descriptor computed inside image patches surrounding SURF interest points. HOG features are classified using a SVM identifying genuine vs fake banknotes. Banknotes were captured

[1] https://legal-dictionary.thefreedictionary.com/

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2020 25th International Conference on Pattern Recognition (ICPR) | 978-1-7281-8808-9/21/$31.00 ©2021 IEEE | DOI: 10.1109/ICPR48806.2021.9412224

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using a digital camera and pre-processed mainly to reduce the noise and improve their quality. In [13] a system to capture banknote images using built-in camera of a mobile device is proposed. After a pre-processing step to reduce the noise using a median filter, a banknote is divided into four equal blocks with respect to its aspect ratio for which discriminative set of features are extracted. Then, an SVM is employed as a classifier. Lohweg et al. [12] focuses on the analysis of intrinsic features produced by Intaglio. A mobile-device banknote image is divided into multiple ROI (up to 360) and a linear discriminant-analysis classifier relies on statistical moments obtained from Wavelet coefficient histograms defined on each ROI. Berenguel et al. [15] tackle the banknotes counterfeit detection with a deep learning approach. A network based on a siamese and a recurrent comparator architecture with attention mechanism is adapted, learning the difference between genuine and counterfeit images through iterative glimpses (8 small regions of interest). Overall, most of the previous approaches rely on texture features defined on ROI selected manually or randomly.

This paper is in the same vein on deep learning in a non-controlled environment, but our network is different. Mainly, we use a full image as input to our network. In [15] two images (genuine vs counterfeit) are alternatively compared on randomly cropped regions. Furthermore, our model uses residual units for better multi-scale feature extraction allowing good abilities to manage low resolution images. Our main contributions include the following aspects.

- We propose an end-to-end trainable deep model the counterfeit banknote detection task, which differs from the literature by using full banknote image and operations for rich multi-scale feature extraction.
- We present a quantitative analysis (what happens inside our network) of the discriminative regions on banknotes, which on some scales, are similar to the preselected patches used in the state-of-the-art.
- We identify new discriminatory regions for counterfeit banknote detection which could be injected into existing datasets.

The rest of this paper is organised as follows. Section II is the heart of our work where we describe our deep learning network of four blocks of depth (respectively 64, 256, 512, and 512) and based on AdapNet++ [16] and ResNet [17]. Unlike previous works, our network has larger receptive fields for a better features resolution and for efficient feature extraction. In Section III experiments results are provided along with analyses and a comparative study with the state-of-art. Finally, Section IV is devoted to the conclusion and we give some perspectives to our work.

## II. Proposed Method

The main characteristics of a real environment are that they are inherently non-controlled. That is, it presents different disturbances when acquiring an image, such as noise and variation in position, lighting, and background. Therefore the main models [8], [18] proposed to use ROI and merge different texture descriptors.



(e) Image (f) Preprocessing (g) Image (h) Preprocessing

Fig. 1. Banknote sample before (left) and after (right) applying the cropping and up-scaling operations. Image is taken from €10 B1 and €50 D1 dataset [18].

However, these previous models adjust a classifier for each banknote value, making this a not entirely viable solution. Although handcraft methods are more comfortable to configure than deep learning models (i.e., the definition of hyperparameters, amount of data learned), they need a trained classifier for each dataset type (i.e., for each value/side of the banknote). By using our deep learning model, we obtain a robust feature extraction (multi-scale), achieving a better generalization of the task. Our experiments show this generalization (see experimental section), where a unique model manages to classify relatively-well even when we use more than one dataset type simultaneously.

Thus, we propose an end-to-end learning model using a deep learning approach. Our model is tested for each of the banknote values, as well as for all of them simultaneously.

Furthermore, contrary to region-based models, we feed our neural network with a complete image of a banknote. We can see in Section III that to produce the prediction, our model makes the features extraction for different ROI at several scales without user intervention. In other words, the model internally learns by itself, which are the regions that will help it to predict better results.

In this section, we describe our network approach to banknote counterfeiting. First, we introduce the preprocessing used to remove part of the background. Second, how we deal with the inherently unbalanced classes. Next, we describe the structure of our network, and finally, we present a training framework for our model.

### A. Preprocessing Banknotes

To feed our network, we need banknotes samples to be in the close-up, i.e., an image that shows a single object in the foreground. Therefore, we cropp a banknote taking into account only the foreground and discarding any regions that can be considered as background. Then, we apply a rescaling which aims to adjust the cropping region such that the overall information is maximized. In Fig. 1, we show an initial sample (left) and after applying the cropping and up-scaling operations (right).

Note that unlike models that use high-quality, noise-free preselected images (i.e., patches), our model deals with image



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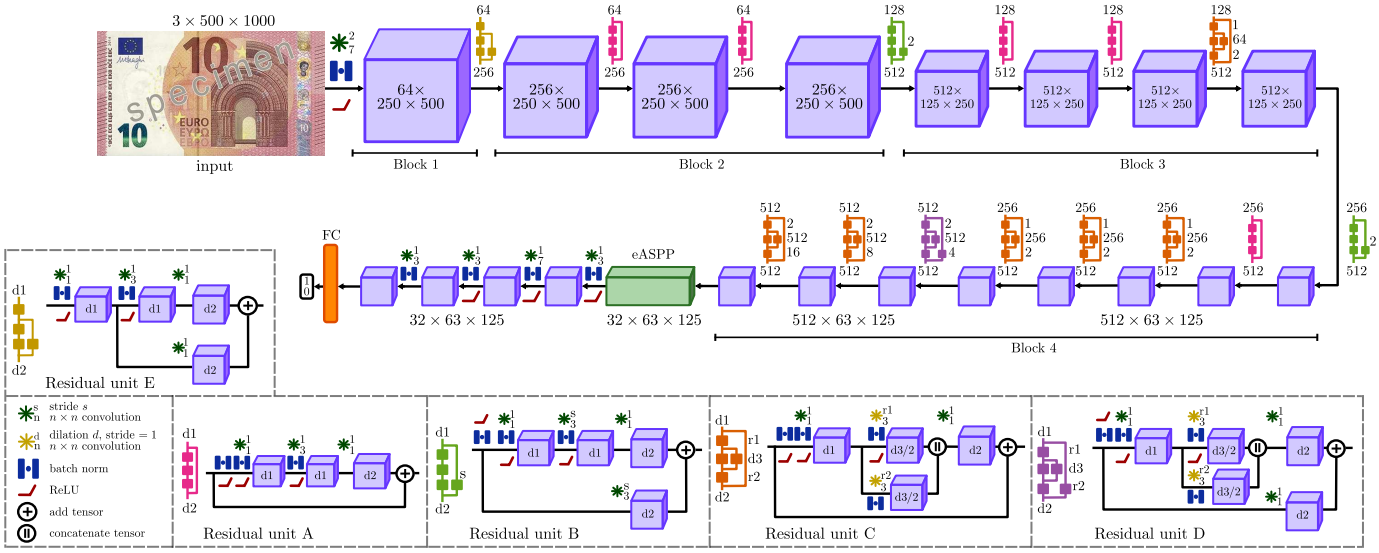



Fig. 2. Our deep learning model, based on AdapNet++ [16] and ResNet [17]. This model consists of four blocks of depth 64, 256, 512 and 512. Besides, it manages the multi-level scales through its different residual blocks ($A$, $B$, $C$, $D$, $E$); it also has larger receptive fields ($eASPP$) for a better features resolution and for efficient feature extraction. Finally, use a fully connected layer to make the banknote counterfeit prediction.

entries of full banknotes. Hand-craft models generally cannot be applied to other datasets, and they are fitted to a single dataset. We show that our model does not have this limitation. Our model has greater generalization but due to the lack of full image banknotes datasets, we use $B3$, $C3$, $D3$, $BCD3$ to show that generalization. Note, the image disturbances (exhibit variations in lighting, position, rotation, shadow, noise) are particularly frequent in €20-C2 (see Table I and Fig 6).

### B. Class Unbalanced

A challenge in the banknote counterfeit detection task is the unbalanced dataset that we have in the training stage [8]. Due to the nature of the data, these always present a lack of counterfeit samples. By having a fewer number of counterfeit samples, the network can fall into overfitting and degenerate the model. To address the unbalanced problem, techniques most used are on the dataset, i.e., resampling (over-sampling and under-sampling), and techniques on loss functions, i.e., class weighted. Because resampling tends to lead to overfitting, we chose to use the class weighted technique. This technique gives higher weight to minority class and lower weight to the majority class. Among these, we selected the median frequency balancing [19] since it gave us the best empirical results. The median frequency balancing is defined by:

$$\tau_c = \frac{median_freq}{freq(c)}, \tag{1}$$

where the $median_freq$ is the median of these frequencies and $freq(c)$ is the number of samples of class $c$ (e.g., positive class) divided by the total number of samples.

In other words, $\tau_c$ is a set hyperparameter (one scalar per class) that increases the penalty of the loss functions (in the training phase) for the classes that have fewer samples.

### C. Network Structure

In recent years, the use of Convolutional Neural Networks (CNNs) has led to several improvements in many fields of knowledge with outstanding results, especially in computational vision. Thus, ResNet [17], [20] architecture achieves good performances on several benchmarks [21], [22]. Fewer parameters and better performance make ResNet become popular in the community. However, we not only need a model with few parameters and faster forward-pass time's, which are critical attributes for deploying models. We also need a model capable of doing a high-resolution feature extraction on the multi-scale level. On the other hand, AdapNet++ [16] proposes a novel technique to efficiently generate multi-scale features throughout the network without increasing the number of parameters. For this reason, we took the encoder of AdapNet++ based on ResNet-50, which meets the previous requirements, and we modified it for the banknote counterfeit detection task.

In the model presented in Fig. 2, we opte to preserve the resolution while reducing the depth of the layers. This is mainly in block 4 and in the Atrous Spatial Pyramid Pooling (eASPP) [16] operations, managing to make the model $\approx 10\%$ slightest. Furthermore, at the end of the network, we add four convolutions of size $3 \times 3$, $7 \times 7$, $3 \times 3$, and $3 \times 3$ to merge

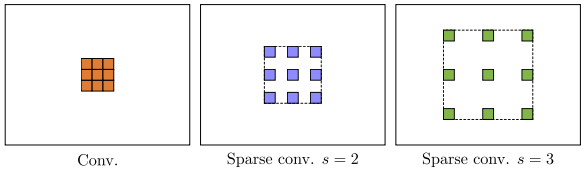



Fig. 3. Graphics of the area visualized by a convolution operation (orange) and by sparse convolutions with different dispersion rates, $s$, (purple and green).



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the features obtained through the eASPP. Finally, we include a fully connected layer (FC) to perform the binary classification.

For better classifier performance, the last four convolutions and the FC, are in charge of processing (by convolution operation) the multi-scale information extracted from the network through the residual units and eASPP.

Note that to capture high-resolution features, we use a larger receptive field (i.e., larger kernel). In other words, we need a wider viewing area in the image. For example, in high-resolution images, using small receptive fields (e.g., orange kernel in Fig. 3) does not provide enough information to classify banknotes. Thus, to achieve the feature extraction in high-resolution, we apply Atrous Convolution [23] (i.e., sparse convolutions), which can have the same parameter number (i.e., $3 \times 3$), but its main characteristics are the dispersion rate parameter s which provides the convolution the capacity to extract features from a larger viewing area (see Fig. 3).

Performing the sparse convolutions is not enough for robust feature extraction in the banknotes. Although, when using sparse convolutions, we obtain a larger area of vision compared to the original convolutions, they still present a single scale vision (e.g., $s = 2$). For this reason, we apply Atrous Spatial Pyramid Pooling (eASPP), which performs sparse convolution operations in parallel with a different rate of dispersion ($s = \{0, 3, 6, 12\}$) without sacrificing efficiency and capturing features in higher resolution (i.e., multi-scale vision).

This way, we keep the higher resolution details while aggregating the same amount of context as before. Furthermore, we want to obtain a rich feature extraction at different levels of scale. This is because, in banknotes, we can obtain different types of information at several scales (see Section III-D). This function is developed by multi-scale units (residual units $A$, $B$, $C$, $D$, $E$ from Fig. 2). These units ($B$, $C$, $D$, $E$), unlike the typical residual unit $A$ used in ResNet, extract, and combine multi-resolution information.

The residual units in Fig. 2 are used in different resolutions (i.e., sample size by downsampling), allowing them to process information at multiple scales, and also to conduct operations with sparse convolutions.

### D. Network Training

Here, we explain the loss function used to fit our deep model. We want to minimize the number of misclassifications on the training dataset, to achieve this, the target function used is a binary cross-entropy, which is defined by:

$$\mathcal{L}_{binary} = -\frac{1}{N}\sum_{c}^{C}\sum_{i}^{N}\tau_c \log P(y_i = c|X;\theta), \qquad (2)$$

where $c$ represents the classes, $y_i$ is the indexed predicted classification (output) for the $i$th class from the set of ground truth $Y$. Additionally, $N$ is the number of classes, and $\theta$ denotes the network parameters to optimize in the stage training. Also, $P(\cdot)$ is the probability that given a banknote image, it is authentic. In this work, we define $P(\cdot)$ by a Sigmoid function:

$$P\left(y_i|X;\theta\right) = \sigma(y_i) \in [0, 1]. \qquad (3)$$

In Eq. 2, each class (genuine and counterfeit banknotes) is weighted by hyperparameter $\tau$. The $\tau$ value is calculated in the preprocessing stage by applying median frequency balancing (see Section II-B). In the counterfeit banknote detection task, fake samples are infrequent. Consequently, $\tau_{c=0}$ of counterfeit class is greater than the $\tau_{c=1}$ of genuine class. This means that the penalty for counterfeit samples misclassification is higher than genuine ones.

TABLE I

EURO DATASET TAKEN FROM [18]. THE ALIAS COLUMN IS ITS SHORT NAME. SIDE REPRESENTS THE WAY THE BANKNOTE WAS PHOTOGRAPHED, OBVERSE($O$), REVERSE ($R$). LIGHT IS THE LIGHTING AND FINALLY OK AND FALSE ARE GENUINE AND COUNTERFEIT BANKNOTES, RESPECTIVELY. THE * SYMBOL DENOTES THE COMBINATION OF DATA TYPES.

| Alias | Banknote | Side | Patch | Ok | False |
|---|---|---|---|---|---|
| $B1$ | 10 € | $O$ | 6 | 346 | 256 |
| $B2$ | 10 € | $R$ | 6 | 348 | 242 |
| $C1$ | 20 € | $O$ | 8 | 392 | 212 |
| $C2$ | 20 € | $R$ | 5 | 373 | 216 |
| $D1$ | 50 € | $O$ | 8 | 350 | 309 |
| $D2$ | 50 € | $R$ | 5 | 351 | 321 |
| $*B3 = B1 + B2$ | 10 € | $O + R$ | – | 694 | 498 |
| $*C3 = C1 + C2$ | 20 € | $O + R$ | – | 765 | 428 |
| $*D3 = D1 + D2$ | 50 € | $O + R$ | – | 701 | 630 |
| $*BCD3 = B3 + C3 + D3$ | 10€ + 20€ +50€ | O+R | – | 2160 | 1556 |

## III. EXPERIMENTS

Very few datasets [15], [24] are available in the litterature. This lack of public ressources is mainly due government data protection rules. For this work, we carry out our experiments on a camera-based Euro banknote dataset [18].

Our experiments were conducted for entire images, unlike traditional methods that use patches (i.e., ROIs). Our model achieves promising results considering that we create a model that classifies all banknotes at the same time, i.e., a single model capable of banknote counterfeit detection independent of the type ($B1$, $B2$, $C1$, $C2$, $D1$, $D2$) or side ($O$, $R$) of the banknote. We also localize the regions used by the model to analyze the results. Next, we first introduce the dataset and the hyperparameters we use in this work. Then, we perform a series of experiments and, we analyze discriminative regions provided by our network on different scales.

### A. Datasets

The dataset [18] contains Euro banknotes by digital and mobile camera-based acquisition. This acquisition of the samples is made with different devices, at several resolutions, and in non-controlled environments. By its nature, banknotes have low lighting, and present noises, e.g., blur or motion blur.

Table I shows more details of the dataset where two different types of lighting are considered. $L_N$ is natural lighting (e.g.,



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TABLE II
BENCHMARK RESULTS, USING THE $\overline{\text{AUC}}$ METRIC ON THE TEST SET. VALUES TAKEN FROM [18].

| | SIFT-BoW | K-SVD | SCSPM | FSIM | 2D-SIWPT | 17Quality | Our Net |
|---|---|---|---|---|---|---|---|
| $B1$ | 0.9646 | 0.9892 | 0.9996 | 0.6718 | 0.7656 | 0.8829 | 0.9964 |
| $B2$ | 0.9606 | 0.9865 | 0.9989 | 0.7766 | 0.7448 | 0.8638 | 0.9676 |
| $C1$ | 0.9859 | 0.9949 | 1.0000 | 0.6664 | 0.7552 | 0.8860 | 0.9762 |
| $C2$ | 0.9822 | 0.9943 | 1.0000 | 0.7105 | 0.7510 | 0.8422 | 0.9111 |
| $D1$ | 0.9799 | 0.9913 | 0.9997 | 0.6399 | 0.6843 | 0.8554 | 0.9968 |
| $D2$ | 0.9643 | 0.9938 | 1.0000 | 0.6477 | 0.6734 | 0.8333 | 0.9774 |
| $*B3$ | – | – | – | – | – | – | 0.9466 |
| $*C3$ | – | – | – | – | – | – | 0.8561 |
| $*D3$ | – | – | – | – | – | – | 0.9975 |
| $*BCD3$ | – | – | – | – | – | – | 0.9612 |

daylight), and $L_A$ is artificial lighting (e.g., a lamp). Image sizes are originally $751 \times 1127$ pixels. Note, after performing the preprocessing (cropping and up-scaling), the final resolution of our input image is $500 \times 1000$ pixels.

### *B. Experimental Setup*

We implement our algorithm using the well-known deep learning framework of Tensorflow. The proposed network is built on AdapNet++. The Atrous convolution is used to have a receptive field of large sizes. The different residual units are used to obtain varied information through various scales. Due to the different number of samples for each type of banknotes, our number of iterations also changes. Models are trained for $616$ iterations for $B1$, $B2$, with $440$ iterations for $C1$, $C2$, and $540$ iterations for $D1$, $D2$. In addition to the combined types of $B3$, $C3$, $D3$, $BCD3$, we use $1424$, $1424$, $1592$, $5570$ iterations, respectively. The model is training using Adam optimizer, with a learning rate of $1e-4$ and batch size $2$ in all the reported experiments. Each input image is cropped and up-scaled to a size of $500 \times 1000$ pixels. Experiments have been conducted on NVIDIA GTX Titan X with $12$ GB of memory.

### *C. Results*

Following the distribution of the dataset, for the evaluation of our model, we divided the samples into $60\%$ for the training phase and $40\%$ for testing ones. In addition, we carry out $k$-fold cross-validation, where we iterate over the dataset set $5$ times. To compare our results, we use the area under the receiver operating characteristic curve (AUC). For each type of dataset and classifiers, AUC is defined by:

$$\overline{\text{AUC}} = \frac{1}{TB} \sum_{b=1}^{B} \sum_{t=1}^{T} \text{AUC}_{t,b}, \tag{4}$$

where $B$ is the total number of partitions in the dataset (5-fold), and $T$ is the total number of texture patches. Note that our model uses $T = 1$, this is because we use a unique integer image that can be interpreted as one patch. Furthermore, $\text{AUC}_{t,b}$ is given by AUC value for the $t$ texture and $b$-sampling (i.e., in $b$-fold).

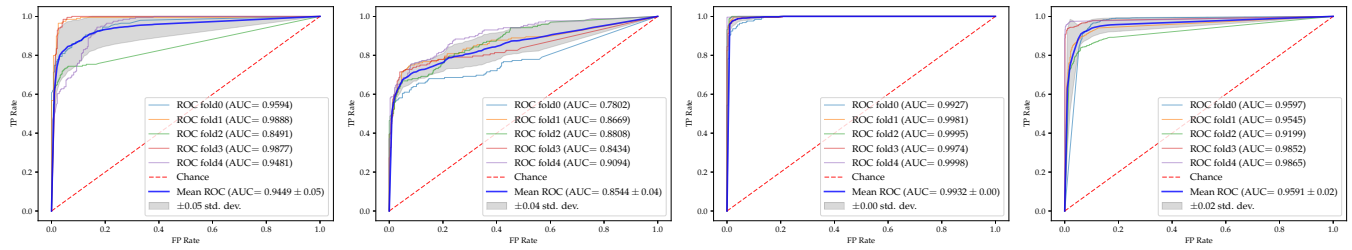
Table II shows a comparison using the $\overline{\text{AUC}}$ metric of our model with the benchmarks (6 different texture classifiers). They use diverse amounts of patches (shown in the fourth column in Table I) to obtain a better likelihood of the predictions. However, although these patches are focused on zones with Intaglio printing [25] (ink that is below the surface used to engrave characters and structures), they are manually chosen, and they must be of high resolution. In addition, the number and the position of these patches is different for each type of dataset (see Table I and in the rightmost of Fig. 5), making these methods not entirely viable in real applications. Contrary to the previous classifiers, which use preselected patches to make the prediction, we input a whole image into our deep neural network and we let it extract the most relevant features at different levels of scales. Thus, by not relying on patches, we can create a banknote counterfeit detection model regardless of the type and side of the banknote (e.g., $B3$, $C3$, $D3$, $BDC3$). Table II shows these results. Combining all the dataset types ($BCD3$ in this table) achieves promising results considering that we don't use zones with Intaglio printing (i.e., patches), and giving way to systems that do not entirely depend on them.

To better analyze the generalizability of our classifier, we plotted the Receiver Operating Characteristic (ROC) curves (Fig. 4), for databases that combine the different types of banknotes (i.e., $B3$, $C3$, $D3$, $BCD3$). We can see the behavior (classification) of our network for the various folds of the dataset in the graphics. Here we can highlight that our model presents a low variance due to the good use of the parameters (i.e., our network appropriately uses the trainable variables to achieve a robust feature extraction). Besides, our network possesses a high increasing rate for TP (our model predicts positive class correctly). Fig. 4 shows that the $C3$ database (from $C1 + C2$) is especially challenging. Note, the FP (our model incorrectly predicts positive class) appears very early in the ROC curve. We attribute this behavior to noise and external components (e.g., shadow) that reduce feature extraction efficiency. In addition, in Table III, we demonstrate the stable behavior of our model by presenting a relatively high response to the different classification metrics. A more detailed analysis (recall, F1-score from Table III) shows that particularly for class $C3$, our model tends to classify false banknotes as originals. This behavior can be corrected by better capturing the input image. The statement is supported by the results of the other data types ($B3$, $D3$) by having a direct relationship between outcomes and sample purity (i.e., images without external components, rotation, motion, noise, shadow).

On the other hand, real-time applications have restricted deadlines, not only in runtime but also in used memory. By reducing the number of parameters used by our deep neural network, we decrease the memory usage required. In Table IV, we present a comparison (in seconds) of the test set and the different benchmarks. We can note that although the best results are obtained by SCSPM, the processing time of this method is high, making it rather unfeasible in a real-time framework. SIFT-BoW and K-SVD models perform comparable results near SCSPM, with a reasonable processing



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(a) $B3$ (b) $C3$ (c) $D3$ (d) $BCD3$

Fig. 4. Graphics of the Receiver Operating Characteristic (ROC) curves on the different types of data combined (B3, C3, D3, BCD3) to visualize the behavior (generalization) of our neural network model. Note, the y-axis represents the true positive rate (TPR), and the x-axis denotes the false positive rate (FPR).

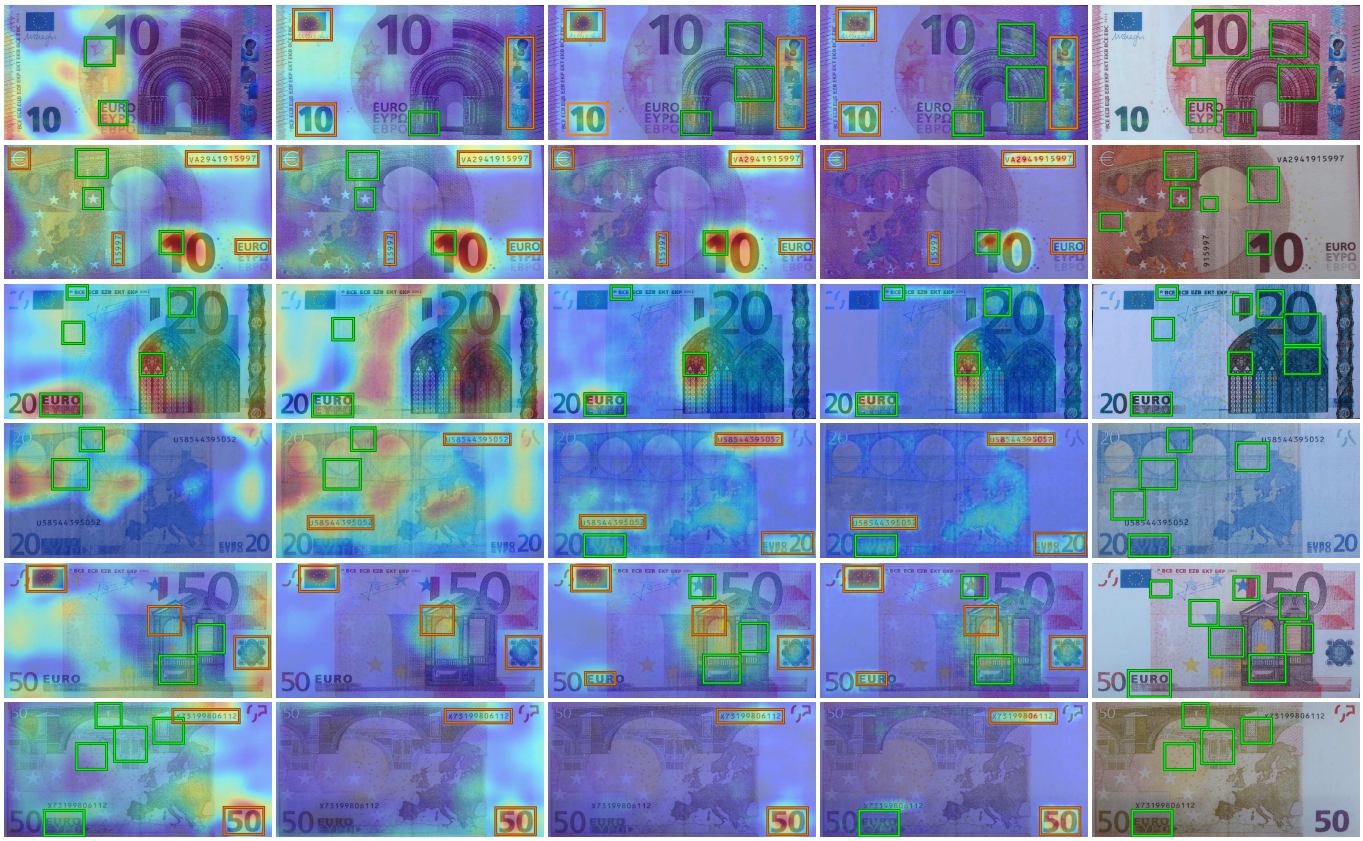


Fig. 5. Discriminative localizations at several scales, obtained by Grad-CAM [26], which were responsible for CNN's prediction. The columns show the regions activated to perform the prediction, on different scales from left to right (eASPP, at the end of block 4, at the beginning of block 4 and at the end of block 3). Rightmost column shows the benchmark patches used by the models.

speed for images of size $256 \times 256$. The performance of our model is between SIFT-BoW and K-SVD, and unlike them, we do not need predefined patches. The execution time, i.e., the forward with a batch size of 2, is small, considering that we use images of size $500 \times 1000$ (see Table IV).

### *D. Analysis*

Current models shed little light on explaining why and how they achieve good performance and, as a result, are treated as black boxes. Therefore, it is essential to understand what these networks learn in order to gain insights into their representations. One way to understand CNNs is to look at the important image regions that influence their predictions.

As we explained previously, in order for our model to manage the features extraction at multiple levels of scale, we use the different residual units. Thus in Fig. 5, we present the activation regions used when making the prediction. Fig. 5 is distributed by data types (the rows) and by scale levels (the columns), where the scales are from left to right obtained respectively by eASPP, at the end of block 4, at the beginning



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TABLE III
COMPLEMENTARY RESULTS ON OUR NEURAL NETWORK MODEL, USING AUC., PRECISION (PREC.), RECALL (REC.), F1-SCORE (F1.), AND ACCURACY (ACC.), ON THE TEST SET.

| | **AUC** $\pm 10^{-3}$ | **Prec** $\pm 10^{-3}$ | **Rec** $\pm 10^{-3}$ | **F1** $\pm 10^{-3}$ | **Acc** $\pm 10^{-3}$ |
|---|---|---|---|---|---|
| $B1$ | $0.997 \pm 0.05$ | $0.996 \pm 0.02$ | $0.994 \pm 0.02$ | $0.995 \pm 0.01$ | $0.996 \pm 0.01$ |
| $B2$ | $0.968 \pm 2.13$ | $0.951 \pm 0.51$ | $0.880 \pm 3.47$ | $0.912 \pm 0.61$ | $0.931 \pm 0.30$ |
| $C1$ | $0.976 \pm 15.4$ | $0.918 \pm 1.87$ | $0.899 \pm 2.30$ | $0.906 \pm 0.16$ | $0.935 \pm 0.43$ |
| $C2$ | $0.911 \pm 33.2$ | $0.920 \pm 0.68$ | $0.887 \pm 1.13$ | $0.903 \pm 0.33$ | $0.930 \pm 0.16$ |
| $D1$ | $0.997 \pm 1.42$ | $0.948 \pm 1.95$ | $0.985 \pm 0.14$ | $0.966 \pm 0.39$ | $0.967 \pm 0.42$ |
| $D2$ | $0.977 \pm 20.3$ | $0.938 \pm 0.44$ | $0.955 \pm 1.67$ | $0.946 \pm 0.87$ | $0.949 \pm 0.75$ |
| $*B3$ | $0.947 \pm 51.3$ | $0.903 \pm 2.79$ | $0.860 \pm 11.9$ | $0.876 \pm 2.87$ | $0.901 \pm 1.55$ |
| $*C3$ | $0.856 \pm 43.6$ | $0.921 \pm 0.89$ | $0.633 \pm 4.64$ | $0.747 \pm 1.56$ | $0.847 \pm 0.24$ |
| $*D3$ | $0.998 \pm 2.56$ | $0.972 \pm 0.53$ | $0.956 \pm 4.33$ | $0.963 \pm 1.04$ | $0.966 \pm 0.76$ |
| $*BCD3$ | $0.961 \pm 24.3$ | $0.957 \pm 1.94$ | $0.849 \pm 18.9$ | $0.891 \pm 6,61$ | $0.919 \pm 2.67$ |

TABLE IV
TIME COMPARISON IN SECONDS ON THE TEST SET. VALUES TAKEN FROM [18].

| | **SIFT-BoW** | **K-SVD** | **SCSPM** | **FSIM** | **2D-SIWPT** | **17Quality** | **Our Net** |
|---|---|---|---|---|---|---|---|
| $256 \times 256$ | 0.1063 | 1.2408 | 2.0997 | 0.2005 | 0.1895 | 0.9072 | – |
| $512 \times 512$ | 0.5265 | 3.5376 | 9.4853 | 0.2317 | 0.4985 | 3.5376 | – |
| $500 \times 1000$ | – | – | – | – | – | – | 0.1159 |



Fig. 6. Examples of poor performance of our model for dataset type $C2$ due to the frequency of noise, such as background regions, the presence of shadows, or poor reference points in bounding boxes. Image is taken from €20 $C2$ dataset [18].

of block 4 and at the end of block 3. The last column shows the different benchmark patches (in green bounding box) used by each type of dataset on texture methods. We observed, in rightmost column, that the position and number of patches change for each type of dataset. Moreover, in the first and second columns, the resolution of the samples is reduced due to the successive convolution operations. Here, our network focuses on more general regions (i.e. large vision field, see the heatmap).

Our models use the regions highlighted with heatmaps to predict counterfeit banknotes. Additionally, we show empirically green and orange bounding boxes, across all scales, to denote the additional regions used by our model (in orange) in addition to using most of the preselected benchmark patches (in green). Our model performs a dynamic selection of the activation regions for different samples of banknotes. Note that certain regions (in orange) remain constant across all scales and adding these regions could improve the texture models.

Finally, we can remark that the poor performance of our model for dataset type $C2$ is due to the exaggerated presence of noise. Fig. 6 shows some preprocessed images with background regions, poor reference points in bounding boxes (i.e., guides in the dataset), blurred noise and, the presence of shadows. This noise problem can be rectified in the acquisition of the database or using static reference points in the bounding box during the camera-based acquisition.

## IV. CONCLUSION

We present an end-to-end trainable model based on Adapnet++, which manages feature extraction at multiple scale levels using several residual units. We solve the problem of unbalance inherent of banknotes datasets by using the median frequency balance on the loss function, thus providing a higher penalty on counterfeit samples. Unlike benchmarks that work with textures on predefined regions of banknotes (i.e., patches), we present an independent model of these, and internally we show that our network uses similar regions or close at different scales, to perform counterfeit banknote detection. Besides, new regions used to make the detection are visualized. We believe that it would be beneficial to add these as patches for the previous models. Our network has been efficiently designed achieving an acceptable prediction time also, due to the reduction of parameters required by our model. Results with the state-of-the-art are competitive, taking into account that we use a simple input image (i.e., a single patch) and without the high resolution required by previous models. Furthermore, our deep neural network shows good abilities when reporting results combining all the different types of datasets.

Finally, we believe that this work can be applied to other kind of documents (for instance identity and/or administrative one) and will guide future research in the search for real-time models with a higher generalization capacity.

## ACKNOWLEDGMENT

This work was financed in part by the São Paulo Research Foundation (FAPESP) under grants No. 2019/18678-0 and, LORIA institute for the infrastructure provided to this research project. We acknowledge the support of NVIDIA Corporation for the donation of a Titan X Pascal GPU used in this research.

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